

Energy Topology, Action, and the Homoclinic Boundary of the Duffing Flow

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29 August 2026 (revision 0)

Abstract

We close an all-parameter phase-space atlas for the conservative Duffing oscillator $\dot{x} = v$, $\dot{v} = -\delta x - \beta x^3$ with $\beta > 0$. The energy levels are classified for both the single-well and double-well regimes, including the two homoclinic loops at the barrier. Turning points are roots of an explicit quadratic in x^2 ; an endpoint-cancelled integral therefore gives the period and action on every regular oval, with the action derivative identity and center, quartic, and saddle limits. The ledger keeps the $\beta = 0$ faces separate and treats the continuum of energy ovals as a source Hamiltonian family, not as isolated primitive cycles. Exact symbolic, independent numerical, replay, and hostile-mutation checks accompany the proof.

1 Model and invariant

Put

$$\dot{x} = v, \quad \dot{v} = -\delta x - \beta x^3, \quad H(x, v) = \frac{v^2}{2} + V(x), \quad V(x) = \frac{\delta x^2}{2} + \frac{\beta x^4}{4}, \quad (1)$$

where $\beta > 0$ and $\delta \in \mathbb{R}$. The physical time is the common clock. Hamiltonian conservation follows immediately from $H_x \dot{x} + H_v \dot{v} = 0$. For a regular connected oval with turning points $\ell < r$, define

$$T(E) = \sqrt{2} \int_{\ell}^r \frac{dx}{\sqrt{E - V(x)}}, \quad I(E) = \frac{1}{\pi} \int_{\ell}^r \sqrt{2(E - V(x))} dx. \quad (2)$$

The endpoint square roots are integrable; the executable ledger evaluates (2) after $x = (\ell + r)/2 + (r - \ell) \sin \theta/2$.

Theorem 1 (complete Duffing energy atlas). *Let $\beta > 0$. If $\delta \geq 0$, every $E > 0$ is one smooth compact oval around the origin. If $\delta < 0$, write $a = \sqrt{-\delta/\beta}$ and $V_{\min} = -\delta^2/(4\beta)$. Then $V_{\min} < E < 0$ gives two ovals, $E = 0$ gives two homoclinic loops to $(0, 0)$, and $E > 0$ gives one outer oval. The squared turning points are*

$$y_{\pm} = \frac{-\delta \pm \sqrt{\delta^2 + 4\beta E}}{\beta}, \quad y = x^2. \quad (3)$$

On each regular component, (2) is finite, $I'(E) = T(E)/(2\pi)$, and the period is positive. For $\delta = -\alpha^2 < 0$, the separatrices are

$$x_h(t) = \pm \frac{\sqrt{2}\alpha}{\sqrt{\beta}} \operatorname{sech}(\alpha t), \quad v_h = \dot{x}_h. \quad (4)$$

Moreover $T(E) \rightarrow 2\pi/\sqrt{\delta}$ as $E \downarrow 0$ when $\delta > 0$, $T(E) E^{1/4} \beta^{1/4}$ is constant when $\delta = 0$, and $T(E)$ diverges logarithmically as a regular energy approaches the saddle level.

Proof. The quartic potential is coercive, so regular compact components are the connected components described by the signs of the two roots in (3). For $\delta < 0$ the identity

$$V(x) + \frac{\delta^2}{4\beta} = \frac{\beta}{4} \left(x^2 + \frac{\delta}{\beta} \right)^2$$

locates the two minima and the barrier. Separation of variables gives (2); the sine substitution removes endpoint singularities. Differentiating the action on a compact subinterval of regular energies and using the vanishing endpoint integrand gives $I' = T/(2\pi)$. At $\delta = 0$, the scaling $x = E^{1/4}\beta^{-1/4}u$ proves the quartic law. The quadratic Taylor expansion at a nondegenerate center gives the center limit. Near the hyperbolic saddle, $E - V(x)$ has a nondegenerate quadratic zero, and comparison with $\int dx/\sqrt{\alpha^2 x^2 + |E|}$ gives the logarithmic divergence. Finally direct substitution of (4) into $\ddot{x} - \alpha^2 x + \beta x^3 = 0$ proves the separatrix formula. $\square$

2 Linear faces and source boundaries

The origin has linear rates $\pm i\sqrt{\delta}$ for $\delta > 0$ and $\pm\sqrt{-\delta}$ for $\delta < 0$; the well equilibria $x = \pm a$ have rates $\pm i\sqrt{-2\delta}$. When $\beta = 0$, the positive- δ face is the harmonic oscillator, the negative- δ face is inverted and has no compact positive-energy ovals, and $(\beta, \delta) = (0, 0)$ is free motion. These lower-dimensional models are not silently folded into the quartic theorem. The two inner components and the outer component are separated in the ledger, so a period near the barrier cannot be mistaken for a center period.

Source note and declarations

The phase-plane terminology follows Guckenheimer and Holmes [1]; period-function conventions follow Chicone [2]. The name Duffing is historical [3]; no priority claim is made. No prime or zero table, local arithmetic datum, Euler factor, root number, automorphy object, target functional equation, Hilbert–Pólya operator or Route-B input is used. Generative tools assisted drafting and code generation; the displayed claims are tied to the artifact chain. This is not external peer review. Scope literal: NO_BAD_EULER_OR_ROOT_NUMBER.

References

- [1] J. Guckenheimer and P. Holmes, *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields*, Springer, 1983. DOI: 10.1007/978-1-4612-1140-2.
- [2] C. Chicone, *Ordinary Differential Equations with Applications*, 2nd ed., Springer, 2006. DOI: 10.1007/0-387-35794-7.
- [3] G. Duffing, *Erzwungene Schwingungen bei veränderlicher Eigenfrequenz*, Vieweg, Braunschweig, 1918.